we've come upon. This doesn't mean that travel to the past is possible—that's a separate issue we'll consider shortly—but it does show that travel back in time can't be ruled out merely by invoking the puzzles we've just discussed.

Rethinking the Puzzles

Recall that in Chapter 5 we discussed the flow of time, from the perspective of classical physics, and came upon an image that differs substantially from our intuitive picture. Careful thought led us to envision spacetime as a block of ice with every moment forever frozen in place, as opposed to the familiar image of time as a river sweeping us forward from one moment to the next. These frozen moments are grouped into *nows*—into events that happen at the same time—in different ways by observers in different states of motion. And to accommodate this flexibility of slicing the spacetime block into different notions of now, we also invoked an equivalent metaphor in which spacetime is viewed as a loaf of bread that can be sliced at different angles.

But regardless of the metaphor, Chapter 5's lesson is that moments—the events making up the spacetime loaf—just are. They are timeless. Each moment—each event or happening—exists, just as each point in space exists. Moments don't momentarily come to life when illuminated by the "spotlight" of an observer's present; that image aligns well with our intuition but fails to stand up to logical analysis. Instead, once illuminated, always illuminated. Moments don't change. Moments are. Being illuminated is simply one of the many unchanging features that constitute a moment. This is particularly evident from the insightful though imaginary perspective of Figure 5.1, in which all events making up the history of the universe are on view; they are all there, static and unchanging. Different observers don't agree on which of the events happen at the same time—they time-slice the spacetime loaf at different angles—but the total loaf and its constituent events are universal, literally.

Quantum mechanics offers certain modifications to this classical perspective on time. For example, we saw in Chapter 12 that on